DESIGN AND FABRICATION OF A 6 DOF PICK AND PLACE ROBOT

A PROJECT REPORT

MARCH 2026

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NACHIMUTHU POLYTECHNIC COLLEGE

POLLACHI-642 003

Government Aided * Autonomous Institution * Approved by AICTE, New Delhi

Affiliated to State Board of Technical Education & Training, Tamilnadu.

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BONAFIDE CERTIFICATE

Certified that this Project report entitled

"Design and Fabrication of a 6 DOF Pick and Place Robot "

Is the Bonafide record work carried out by

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We pay our sincere and honorable salutation to the almighty for equipping us with all the strength and courage throughout our venture in the project.

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CHAPTER – 1

ABSTRACT

Our project "Design and Fabrication of A 6 DOF Pick and Place Robot" aims to develop an advanced robotic arm system capable of performing precise and controlled movements for handling objects. The robotic arm is designed using servo motors and a microcontroller-based control system to achieve smooth and accurate motion. Our system uses different electronic and mechanical components, and its various operations are controlled by embedded software. It offers a flexible and efficient platform for demonstrating robotic motion, object manipulation, and automation concepts. The system improves understanding of robotic joint coordination, motion control, and speed regulation. It also serves as a valuable tool for studying robotic behaviour, control techniques, and enhancing practical knowledge in automation and embedded systems at reduced cost.

The existing industrial robotic arms available in the market are very costly and complex; hence we proposed to use affordable electronic components and a simplified mechanical structure to develop the system.

This robotic arm helps to understand robotic mechanisms in a shorter period of time with hands-on experience in motion control and programming. Our robotic arm consists of the following components:

- ESP32 Microcontroller
- Servo Motors for 6 axis
- Buck Converter Power Supply
- Mechanical Aluminium Structure
- Capacitors for Voltage Stabilization
- Wireless Web-Based Control Interface

All the above components are assembled and integrated with suitable programming software to ensure smooth, stable, and efficient robotic operation.

CHAPTER – 2

INTRODUCTION

In the modern era of technology, automation and robotics are playing a crucial role in transforming industries, research fields, and educational systems. Machines are increasingly being designed to perform tasks that require precision, speed, and consistency. Among these technological advancements, the robotic arm stands out as an important innovation that replicates the motion and functionality of a human arm. A robotic arm combines mechanical structure, electronic components, and embedded programming to perform controlled and accurate movements.

Robotic arms are widely used in industrial environments for applications such as welding, assembling, painting, packaging, and material handling. These systems improve productivity, reduce human effort, and enhance workplace safety by operating in hazardous or repetitive task environments. However, industrial robotic systems are often expensive and complex, making them less accessible for academic institutions and small-scale applications.

To overcome these limitations, low-cost and compact robotic arms controlled by microcontrollers have been developed. In this project, a 6 Degrees of Freedom (6 DOF) robotic arm is designed and fabricated using servo motors and an ESP32 microcontroller. The servo motors provide precise angular control of each joint, while the ESP32 generates PWM signals to control position and speed. The integration of a wireless web-based interface enables real-time control of the robotic arm using a mobile phone or computer.

The system allows users to move individual joints, adjust speed, record motion sequences, and replay tasks. This enhances learning by providing hands-on experience in robotic kinematics, motion coordination, and embedded system programming. We can understand how mechanical design and software control work together to achieve smooth and stable movement.

Additionally, proper power regulation and voltage stabilization techniques are implemented to ensure reliable performance of multiple servo motors. The robotic arm is designed to maintain a home position at startup and provide smooth movement to avoid sudden jerks or instability.

This project not only demonstrates the fabrication of a functional robotic manipulator but also highlights the practical application of automation principles. It serves as an educational platform for understanding robotics, control systems, and wireless communication. As automation continues to grow across industries, knowledge and experience in robotic systems will become increasingly valuable. The development of this robotic arm represents a step toward mastering modern robotic and automation technologies.

CHAPTER – 3

LITERATURE SURVEY

3.1 Robotic Arm Literature Survey: A Review of Key Research and Developments

Robotic arms have emerged as one of the most transformative technologies in the field of automation and robotics. According to John J. Craig (2005) in *Introduction to Robotics: Mechanics and Control*, robotic manipulators are programmable mechanical devices designed to move materials, parts, tools, or specialized devices through variable programmed motions for the performance of a variety of tasks. Over the past several decades, researchers have focused on improving precision, efficiency, and flexibility in robotic systems.

M. P. Groover (2015) emphasized that industrial robotic arms have significantly increased manufacturing productivity by reducing human error and enhancing repeatability. Early robotic systems were primarily hydraulic and pneumatic, designed for heavy industrial applications. These systems were powerful but lacked precision and required extensive maintenance.

With advancements in electronics and embedded systems, electric servo motor–based robotic arms became dominant. According to Siciliano and Khatib (2016) in *Springer Handbook of Robotics*, electric actuators provide better control accuracy, reduced noise, and improved energy efficiency compared to hydraulic systems. The integration of microcontrollers enabled precise motion control through PWM techniques, making robotic arms more adaptable and programmable.

Recent studies have shifted toward low-cost robotic arm development. Researchers such as Kumar and Rao (2019) proposed microcontroller-based robotic arms using Arduino platforms for educational purposes. Similarly, ESP32-based robotic systems have gained attention due to built-in Wi-Fi capabilities, enabling wireless monitoring and remote control.

3.2 Pioneers and Early Developments

The foundation of modern industrial robotics began with George Devol and Joseph Engelberger in 1961, who introduced the first industrial robot, Unimate. This robotic arm was used in automotive assembly lines for die-casting and welding operations. Their work marked the beginning of programmable automation.

Victor Scheinman (1969) developed the Stanford Arm, one of the first electrically powered robotic manipulators controlled by a computer. His design introduced joint-based control and programmable positioning, which became a standard model for future robotic arms.

Research in robotic kinematics was significantly advanced by Denavit and Hartenberg (1955), who introduced the Denavit–Hartenberg (D-H) parameter method. This mathematical representation simplified the modeling of robotic joint movements and is still widely used for forward and inverse kinematics calculations.

Paul (1981) further contributed to the field by developing algorithms for robotic motion planning and trajectory generation. These developments improved path accuracy and minimized positioning errors.

Over time, research expanded into areas such as adaptive control, feedback systems, and dynamic modeling, allowing robotic systems to perform increasingly complex tasks.

3.3 Research in Motion Control and Embedded Systems

The evolution of robotic arms closely follows developments in embedded systems and control theory. Ogata (2010) highlighted the importance of PID control systems in achieving stable and smooth actuator motion. Many robotic arm systems use proportional–integral–derivative control for accurate joint positioning.

Servo motor technology has also evolved significantly. According to Bolton (2015), servo motors provide closed-loop feedback systems, ensuring high precision and stability. Their compact size and affordability make them suitable for small-scale robotic manipulators.

Research by Zhang et al. (2020) demonstrated the implementation of Wi-Fi–enabled robotic arms using ESP32 microcontrollers. Their study emphasized real-time web control, motion recording, and remote monitoring capabilities. The integration of iot platforms has further expanded robotic arm applications in smart automation systems.

Furthermore, researchers have focused on power stabilization techniques. Voltage fluctuation caused by multiple servo motors can lead to instability. Studies recommend using buck converters and high-capacitance electrolytic capacitors to ensure consistent power supply.

3.4 Applications in Industry and Education

Industrial Automation

Groover (2015) states that robotic arms are widely used in welding, painting, packaging, and material handling. Industrial robots improve efficiency, ensure precision, and reduce human exposure to hazardous environments. Studies show that automation increases production rates and reduces operational cost over time.

Medical and Surgical Robotics

Taylor et al. (2008) explored robotic arms in surgical applications, emphasizing high precision and minimal invasive procedures. Robotic systems in healthcare demonstrate the potential for improved accuracy beyond human limitations.

Educational and Research Platforms

Recent literature highlights the importance of low-cost robotic arms in education. According to Monk (2016), Arduino and ESP32 platforms provide accessible tools for learning embedded systems and robotics. These platforms allow students to experiment with PWM control, kinematics, and automation principles in a practical environment.

IOT and Wireless Robotic Systems

IOT integration has become a major research area. Studies by Kumar et al. (2021) show that wireless robotic systems improve accessibility and enable remote diagnostics. Web-based interfaces allow users to monitor and control robotic arms from anywhere within a network.

3.5 Validation and Performance Analysis

Validation of robotic arm systems focuses on parameters such as:

- Positional accuracy
- Repeatability
- Load capacity
- Response time
- Power efficiency

Researchers emphasize the importance of stable power systems and smooth motion algorithms to reduce mechanical stress and vibration. According to Spong, Hutchinson, and Vidyasagar (2006), precise modeling and feedback control are essential for achieving high-performance robotic systems.

Testing methods typically involve repeated motion cycles to evaluate consistency. Error margins are calculated to determine system accuracy. Studies show that servo-based robotic arms are suitable for lightweight tasks and educational experiments.

3.6 Emerging Trends and Future Research

The future of robotic arms lies in the integration of Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision. According to Russell and Norvig (2016), AI-driven robotics enables autonomous decision-making and object recognition.

Collaborative robots (cobots) are another emerging development. These systems are designed to safely interact with humans in shared workspaces. Advanced sensors and adaptive algorithms ensure safe operation.

Researchers are also exploring lightweight composite materials and energy-efficient actuators to improve robotic performance. AI-enabled robotic arms can adapt to environmental changes and optimize motion paths automatically.

As automation continues to expand across industries, robotic arms will play a critical role in smart manufacturing, healthcare assistance, space exploration, and domestic robotics. The combination of embedded systems, wireless communication, and intelligent algorithms will further enhance the capabilities of robotic manipulators.

CHAPTER – 4

WORKING PRINCIPLE

4.1 Principle of Working of Robotic Arm

The working principle of the robotic arm is based on the coordination between user input, microcontroller processing, and servo motor actuation. The complete process is explained below:

1. User Input

The user interacts with the robotic arm through a web-based control interface hosted by the ESP32 microcontroller. The interface contains sliders, buttons (Record, Play, Home), and speed control options. When the user moves a slider or presses a button, a command is sent wirelessly to the ESP32.

2. Signal Transmission

The control signals are transmitted through Wi-Fi communication between the user's device (mobile/PC) and the ESP32 module. These signals contain angle values and speed parameters for each servo motor.

3. Microcontroller Processing

The ESP32 receives the command and interprets the requested angle positions. Based on the programmed logic, it calculates the required PWM (Pulse Width Modulation) signals to rotate each servo motor to the desired position.

4. PWM Signal Generation

The ESP32 generates PWM signals corresponding to specific servo angles (typically 0° to 180°). The pulse width determines the position of the servo shaft.

5. Servo Motor Actuation

Each servo motor receives its respective PWM signal and rotates to the commanded angle. Since the robotic arm has 6 Degrees of Freedom (Base, Shoulder, Elbow, Wrist Pitch, Wrist Roll, and Gripper), coordinated movement of all servos results in precise arm positioning.

6. Motion Control and Speed Regulation

The system includes a speed control feature that adjusts the delay between incremental angle movements. This ensures smooth motion and prevents sudden jerks, thereby improving stability and reducing mechanical stress.

7. Home Position Initialization

When powered on, the robotic arm automatically moves to a predefined home position. This ensures consistent startup alignment and prevents random or unstable movements.

8. Record and Playback Function

The system can store sequences of servo positions during recording mode. When playback is activated, the ESP32 sequentially re-executes the stored positions, allowing the arm to repeat tasks automatically.

4.1.2 Advantages

• Precise control using PWM

• Wireless web-based operation

• Smooth and adjustable speed control

• Repeatable motion through recording feature

• Cost-effective compared to industrial robots

• Educational and research friendly

4.1.3 Limitations

• Limited payload capacity

• Servo motors may experience jitter under heavy load

• Requires stable power supply for multiple servos

• Limited industrial-grade precision

• Dependent on Wi-Fi connectivity for web control

4.2 WORKING BLOCK DIAGRAM

Below is the working block representation of the robotic arm system.

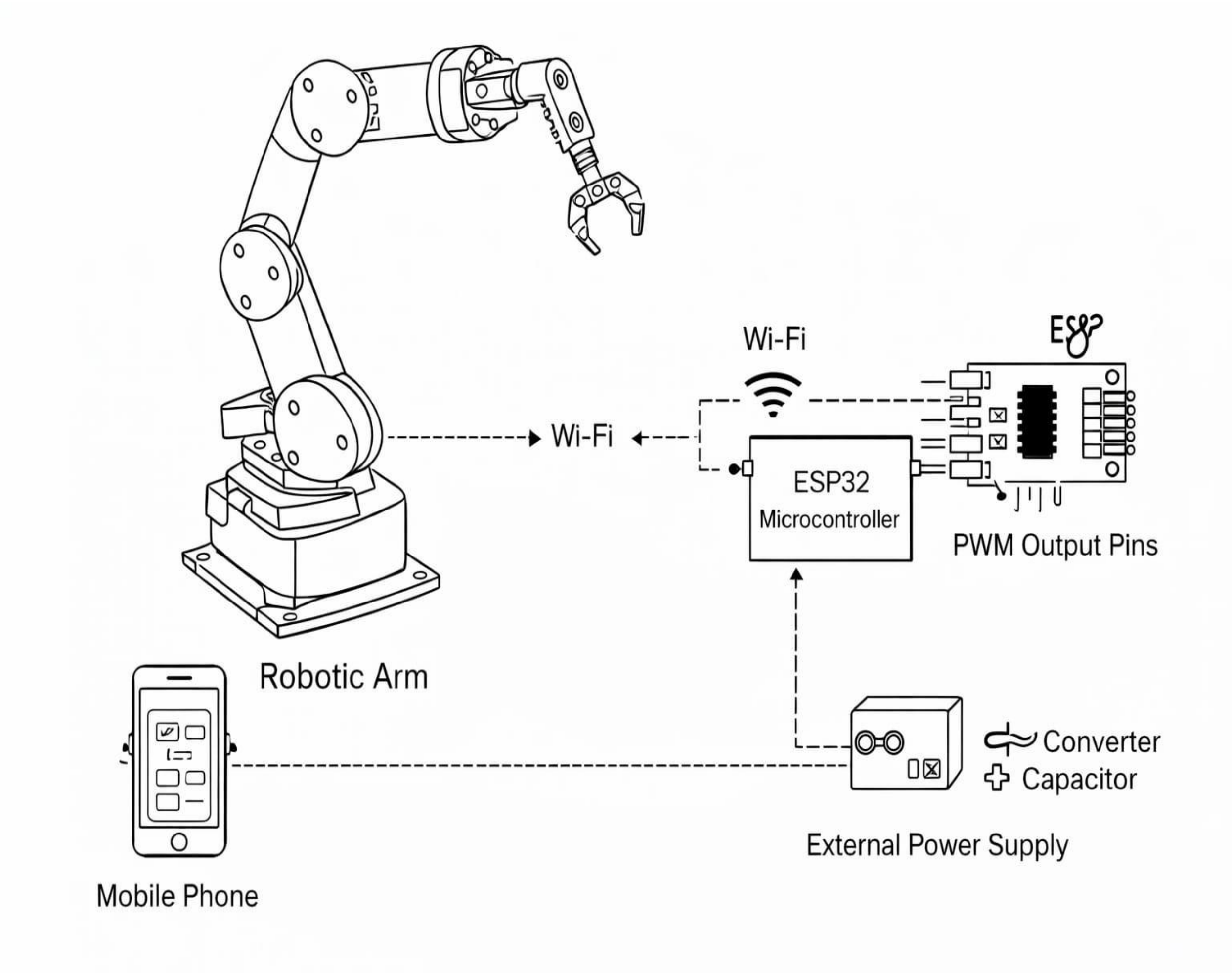


Fig 4.1 – Block Diagram of Robotic Arm System

Block Explanation:

1. User Interface (Mobile/PC Browser) – Website
2. Wi-Fi Communication
3. ESP32 Microcontroller
4. PWM Output Pins
5. Servo Motors (6 DOF) - 6
6. Robotic Arm Mechanical Structure
7. External Power Supply (Buck Converter + Capacitor)

CHAPTER – 5

DESCRIPTION OF EQUIPMENTS

5.1 ESP32 Microcontroller

The ESP32 microcontroller serves as the central processing unit of the robotic arm system. It functions as the brain of the entire setup, coordinating communication, signal processing, motion control, and execution of programmed tasks. The ESP32 is a powerful and versatile microcontroller specifically designed for Internet of Things (IoT) and embedded applications. It integrates built-in Wi-Fi and Bluetooth capabilities, making it highly suitable for wireless automation systems. In this project, the wireless communication feature of the ESP32 plays a crucial role in enabling web-based control of the robotic arm.

The ESP32 operates using a dual-core 32-bit processor, which allows efficient multitasking. One core can handle communication tasks such as hosting the web server and managing Wi-Fi connectivity, while the other core manages real-time servo motor control. This separation of tasks ensures smooth system performance without delay or interruption. The high clock speed of the ESP32 enhances its ability to process commands quickly, resulting in responsive robotic arm movements.

In the implemented system, the ESP32 hosts an internal web server that provides a user-friendly interface for controlling the robotic arm. When the user accesses the web page through a mobile phone or computer connected to the ESP32's Wi-Fi network, control commands are transmitted wirelessly. These commands include servo angle adjustments, speed control modifications, record and playback instructions, and home position activation. The ESP32 continuously listens for these inputs, processes them in real time, and executes the corresponding actions.

One of the major advantages of using the ESP32 is its multiple GPIO (General Purpose Input/Output) pins, which support PWM signal generation. Since the robotic arm consists of six servo motors, each motor requires an independent PWM signal for accurate positioning. The ESP32 efficiently generates these PWM signals with precise timing control. The pulse width determines the angular position of each servo, typically ranging from 0° to 180°. By adjusting the pulse duration, the ESP32 ensures smooth and controlled joint movement.

The microcontroller also manages additional features such as speed regulation, motion recording, and playback functionality. Speed control is implemented by adjusting the delay between incremental servo movements, allowing gradual and stable motion. During recording mode, the ESP32 stores servo positions and timing data in memory. In playback mode, the stored sequence is executed automatically, enabling repetitive task automation.

Power efficiency is another important characteristic of the ESP32. Although it is a high-performance microcontroller, it operates with relatively low power consumption, making it suitable for battery-powered or portable systems. However, since servo motors require higher current, an external regulated power supply is used while maintaining a common ground reference with the ESP32.

The ESP32 supports programming through the Arduino IDE, which simplifies development, debugging, and future upgrades. Its flexibility allows easy integration of additional sensors, cameras, or IoT features if required in future expansions of the project.

Overall, the ESP32 acts as the intelligence unit of the robotic arm system. It ensures accurate movement control, reliable wireless communication, multitasking capability, and smooth execution of programmed operations. Its combination of performance, connectivity, and flexibility makes it an ideal choice for embedded robotic applications.

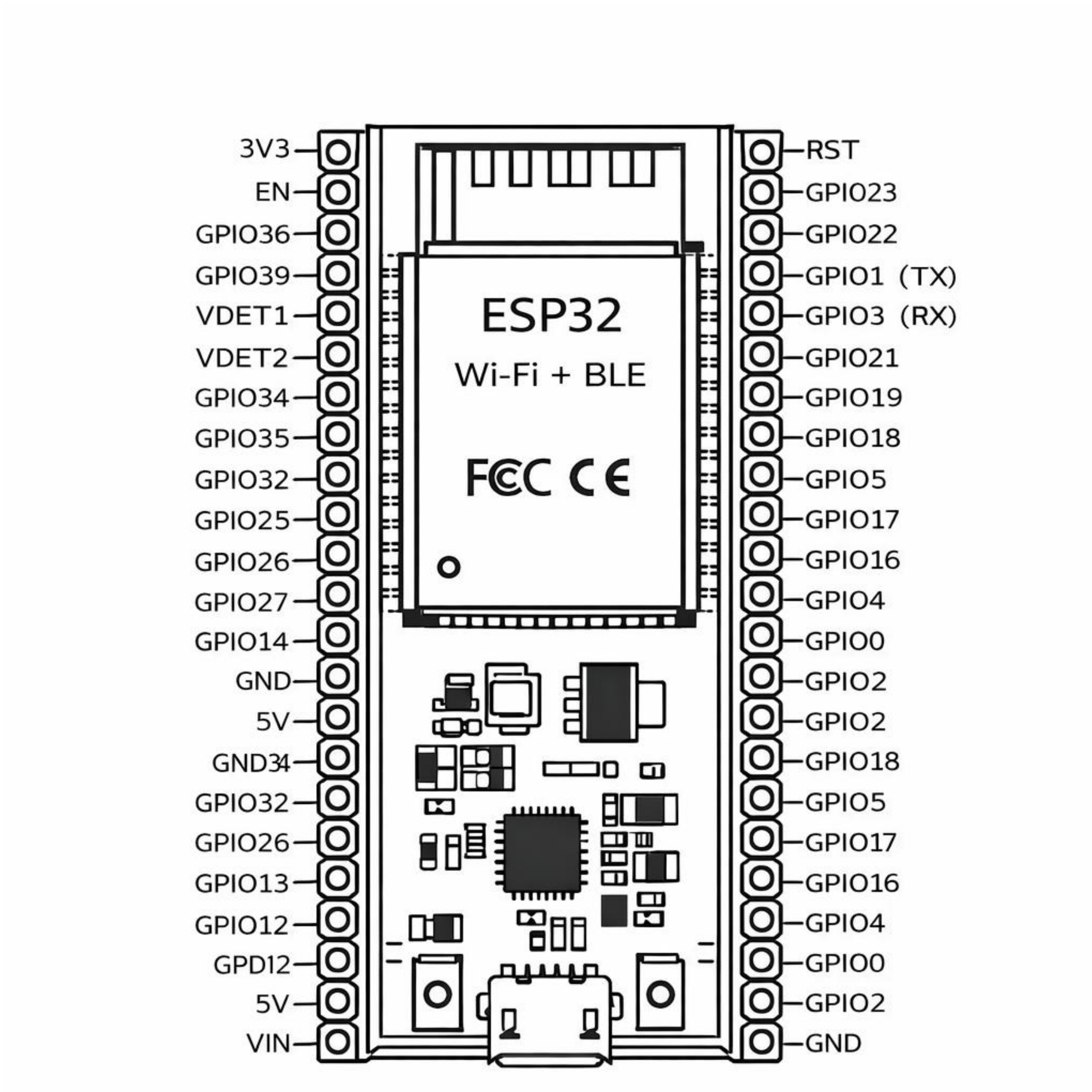


Fig 5.1 – Block Diagram of ESP32 Microcontroller

5.2 Servo Motors

Servo motors are the primary actuators responsible for generating mechanical movement in the robotic arm system. In this project, six servo motors are utilized to achieve six degrees of freedom (6 DOF), enabling the robotic arm to perform multi-directional movements similar to the joints of a human arm. Each servo motor is assigned to a specific joint, namely the base rotation, shoulder movement, elbow movement, wrist pitch, wrist roll, and gripper operation. The coordinated functioning of these six motors allows the robotic arm to execute complex and precise motions.

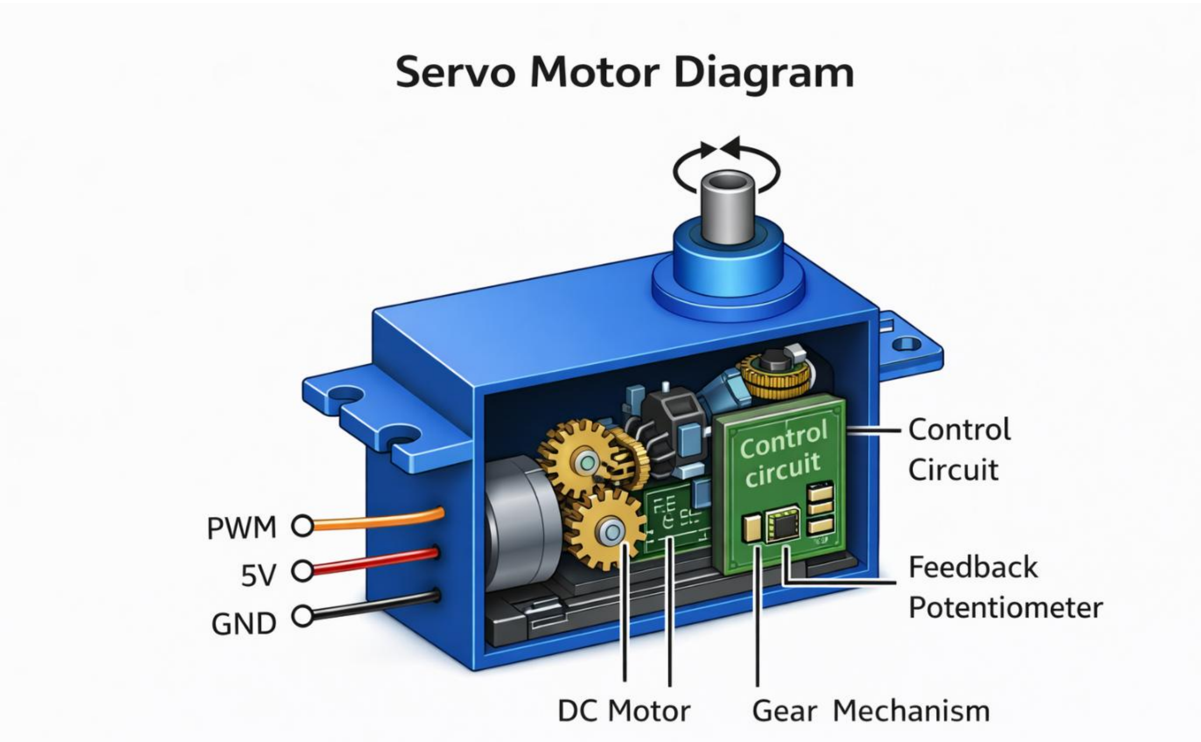
Servo motors operate based on Pulse Width Modulation (PWM) signals generated by the microcontroller. A PWM signal consists of a series of electrical pulses where the width of each pulse determines the angular position of the servo shaft. Typically, standard hobby servo motors rotate within a range of 0° to 180°. By varying the pulse width, the servo motor adjusts its shaft position accordingly. This precise angular control makes servo motors highly suitable for robotic applications where accurate and repeatable positioning is essential.

Internally, a servo motor consists of four major components: a DC motor, a gear reduction mechanism, a control circuit, and a feedback potentiometer. The DC motor provides rotational movement, while the gear system reduces speed and increases torque to handle mechanical loads. The feedback potentiometer continuously monitors the current shaft position and sends this information to the internal control circuit. The control circuit compares the desired position (from the PWM input) with the actual position and corrects any error by adjusting motor movement. This closed-loop feedback mechanism ensures stable, accurate, and vibration-free positioning even when the motor is subjected to external loads.

In this robotic arm project, servo motors are selected due to several advantages. They are compact in size, lightweight, affordable, and easy to interface with microcontrollers such as the ESP32. Additionally, they provide sufficient torque for handling lightweight objects and small mechanical structures. Their simple three-wire connection (power, ground, and signal) makes wiring straightforward and convenient.

However, since multiple servo motors operate simultaneously, they require a stable and adequate power supply. Sudden movements or heavy loads may cause current spikes, leading to voltage drops or jittering. Therefore, a regulated external power source is used to ensure smooth and uninterrupted operation.

By synchronizing the movements of all six servo motors, the robotic arm can perform tasks such as picking, placing, rotating, gripping, and positioning objects with controlled precision. Thus, servo motors play a vital role in converting electrical signals into accurate mechanical motion, forming the core actuation system of the robotic arm.



5.2 Servo Motor

5.3 Power Supply Unit (Buck Converter and Capacitor)

A stable and reliable power supply is essential for the smooth and uninterrupted operation of the robotic arm system. Since servo motors consume a significant amount of current, especially during startup or sudden directional changes, relying solely on the ESP32's onboard power supply is insufficient. Therefore, an external power source is used to ensure adequate voltage and current availability for all servo motors.

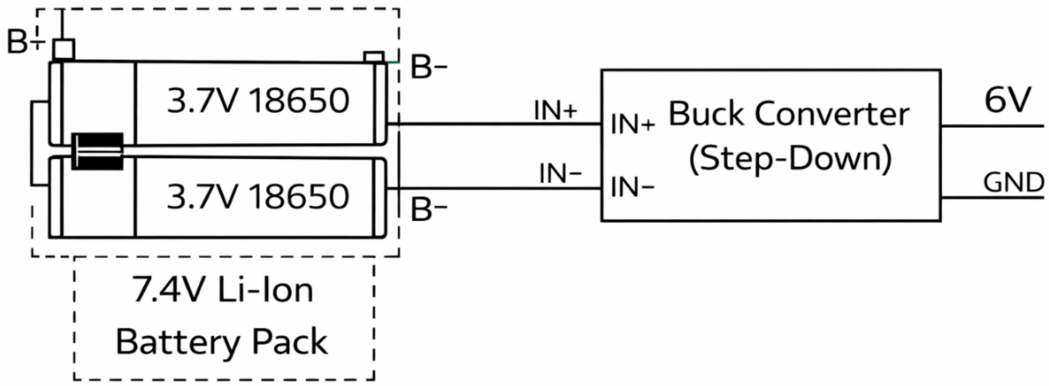
In this project, a battery source provides the required input voltage. However, servo motors typically operate efficiently at around 6V. If a higher voltage source such as 8V–12V is directly supplied to the servos, it may cause overheating, excessive vibration, or permanent damage. To prevent this, a buck converter is used. A buck converter is a DC–DC step-down voltage regulator that converts higher input voltage to a lower, stable output voltage. It operates efficiently by switching and regulating the voltage through internal circuitry, ensuring minimal power loss and heat generation.

The buck converter is adjusted to provide a constant 6V output suitable for driving the servo motors. Voltage regulation is extremely important because insufficient voltage may result in weak torque, jittering, or unstable motion, while excessive voltage may damage internal servo components.

To further stabilize the power supply, a high-capacitance electrolytic capacitor (4700 µF, 25V) is connected across the output terminals of the buck converter. This capacitor acts as an energy storage device and helps reduce voltage fluctuations. When multiple servos start moving simultaneously, sudden current spikes occur. The capacitor absorbs these spikes and supplies additional current momentarily, preventing voltage drops and ensuring consistent performance.

Another critical aspect of the power system is maintaining a common ground between the ESP32 microcontroller and the servo power supply. The ground of the buck converter output is connected to the ESP32 ground to ensure proper signal referencing. Without a common ground connection, PWM signals sent to the servo motors may behave unpredictably, leading to erratic movement or control errors.

The power supply unit therefore plays a vital role in improving system reliability, reducing jittering, preventing vibration, and protecting electronic components from sudden voltage drops. A properly designed power system enhances the overall durability and stability of the robotic arm.



5.3 Battery Pack

5.4 Mechanical Structure of Robotic Arm

The mechanical structure of the robotic arm forms the physical backbone of the entire system. It provides the framework that supports the electronic components, servo motors, wiring, and control units. A well-designed mechanical structure is essential to ensure stability, smooth motion, strength, and durability during operation. In this project, the robotic arm is designed with multiple interconnected joints and rigid links that mimic the structure and movement of a human arm.

The base of the robotic arm acts as the primary support and foundation. It is designed to hold the entire assembly firmly and prevent unwanted vibrations or tilting during operation. The base houses the first servo motor, commonly referred to as the base servo, which provides rotational movement along the horizontal axis. This rotation enables the arm to turn left and right, increasing its working range.

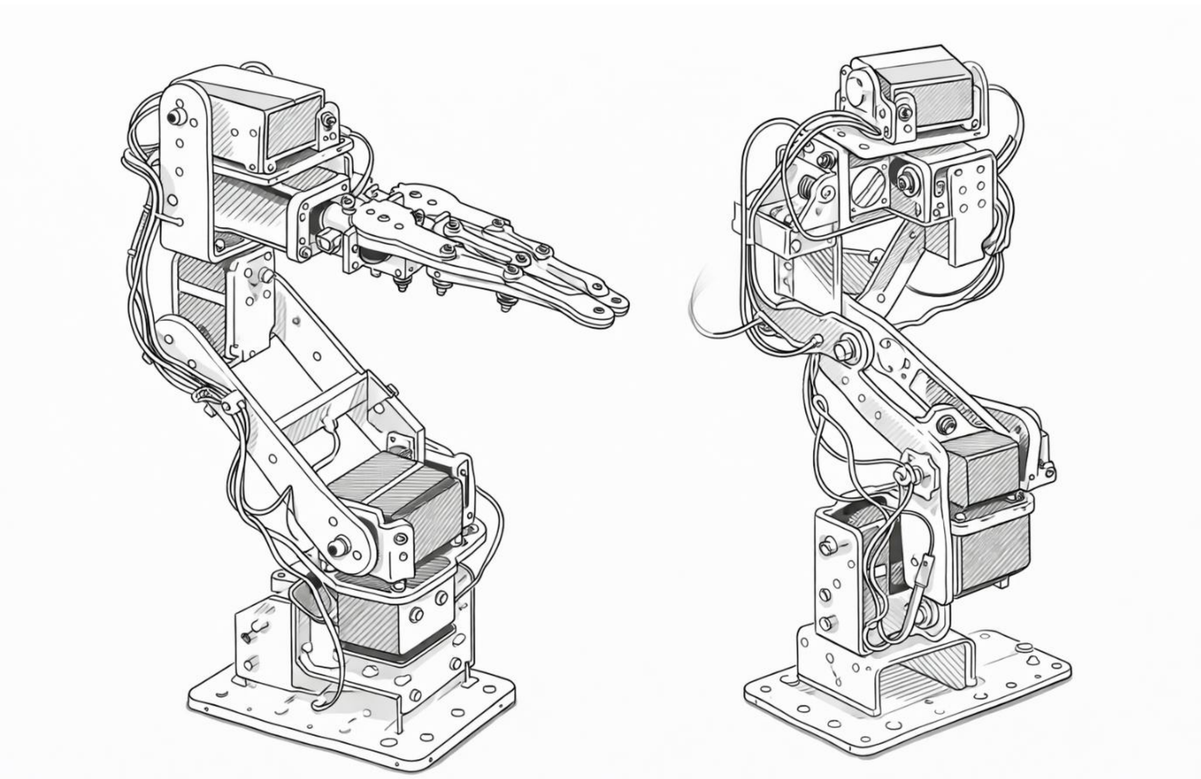
Above the base, the arm consists of several joints connected through mechanical links. These joints form the shoulder, elbow, and wrist sections of the robotic arm. Each joint is powered by an individual servo motor, allowing controlled angular displacement. The shoulder joint enables vertical lifting motion, the elbow joint extends or retracts the arm, and the wrist joints provide additional flexibility for precise positioning. This multi-joint configuration provides the robotic arm with six degrees of freedom (6 DOF), allowing it to perform complex movements in three-dimensional space.

The links connecting the joints are typically made of lightweight yet strong materials such as aluminum, acrylic, or high-strength plastic. Lightweight construction reduces the load on servo motors, improving efficiency and minimizing power consumption. At the same time, sufficient rigidity is maintained to prevent bending or deformation under load.

Proper alignment of servo motors and mechanical parts is critical. Incorrect alignment can cause unnecessary strain, vibration, or torque imbalance, which may reduce accuracy and shorten component lifespan. Secure mounting brackets and firm fastening ensure efficient torque transfer from the servo motor shafts to the joints.

At the end of the arm, a gripper mechanism is installed. The gripper functions as the end-effector, enabling the robotic arm to grasp, hold, and release objects. The gripping mechanism is typically controlled by a dedicated servo motor, allowing adjustable opening and closing movements depending on object size.

Overall, the mechanical structure transforms electrical control signals into controlled physical motion. It ensures precision, stability, and reliability, making it a crucial component in achieving the functional performance of the robotic arm system.



5.4 Mechanical Structure of Robotic Arm

5.5 Web-Based Control Interface

The web-based control interface provides a user-friendly method to operate the robotic arm wirelessly. The ESP32 hosts a local web server that generates a control webpage accessible through a mobile phone or computer connected to the same Wi-Fi network.

Through this website, the user can directly control the **Base rotation, Shoulder movement, Elbow movement, Wrist pitch, Wrist roll, and Gripper operation** of the robotic arm. The interface contains sliders for adjusting the angle of each servo motor individually. It also includes speed adjustment buttons to control the movement speed of the arm, a home position button to return the arm to its predefined starting position, and record and playback functions to automate repetitive tasks.

When the user interacts with these controls, the browser sends angle values and control commands to the ESP32 in real time. The microcontroller processes these inputs and generates corresponding PWM signals to move the servo motors to the selected positions.

One of the major advantages of this system is that it eliminates the need for physical controllers or additional hardware interfaces. The user can control all joint movements and functional operations of the robotic arm remotely through any device with a web browser. This increases flexibility, convenience, and ease of operation.

The record function allows the system to store a sequence of joint movements performed from the website, while the playback function enables automatic repetition of the recorded task. The speed control feature adjusts the delay between incremental servo movements, ensuring smooth, stable, and controlled operation.

Thus, the web-based interface provides complete control over the robotic arm’s motion, positioning, speed regulation, and automation features, making it suitable for educational demonstrations, research experiments, and small-scale automation applications.

CHAPTER – 6

MERITS

1. Wireless control system

The robotic arm is controlled through a web-based interface using WI-FI communication. This eliminates the need for physical controllers and allows remote operation using a mobile phone or computer.

2. Precise motion control

The use of servo motors and PWM signals ensures accurate angular positioning. Each joint can be precisely controlled, allowing smooth and stable movement.

3. Speed adjustment feature

The system includes a speed control mechanism that enables the user to increase or decrease the movement speed of the servo motors. This improves motion smoothness and prevents sudden jerks.

4. Record and playback function

The robotic arm can record movement sequences and replay them automatically. This feature allows repetition of tasks without manual control, improving automation capability.

5. Home position initialization

The arm automatically moves to a predefined home position at startup. This ensures consistent alignment and prevents random motion during power-on.

6. Cost-effective design

Compared to industrial robotic arms, this system is economical and suitable for educational and small-scale applications.

7. Compact and lightweight

The mechanical structure is lightweight, making it portable and easy to assemble.

8. Easy programming and modification

Since the system is programmed using the ARDUINO IDE and ESP32 platform, modifications and improvements can be easily implemented.

9. Expandable system

Additional sensors such as cameras, ultrasonic sensors, or force sensors can be integrated to enhance functionality.

DEMERITS

1. Limited load capacity

The robotic arm uses small servo motors that cannot lift heavy objects. Its payload capacity is limited, restricting it to lightweight tasks such as basic picking and placing operations.

2. Power supply sensitivity

The servo motors require a stable voltage to operate smoothly. Any fluctuation in the power supply can cause jitter, reduced torque, or unexpected movements, affecting reliability.

3. Limited precision compared to industrial robots

Although servo motors provide good control, they cannot match the accuracy and repeatability of high-end industrial robotic arms. Small errors may accumulate during complex movements.

4. Wi-fi range limitation

Since the system depends on WI-FI CONNECTIVITY, control is limited to the signal range of the ESP32 ACCESS POINT. Physical obstacles or interference can reduce responsiveness or disconnect the controller.

5. Mechanical wear and joint stress

Repeated movement over long periods can cause mechanical wear on servo gears, joints, and metal brackets. This may lead to reduced smoothness, noise, or the need for periodic maintenance.

CHAPTER – 7

APPLICATIONS

1. Material handling (light mechanical tasks)

The robotic arm can be used for lifting and transferring small mechanical components such as bolts, nuts, washers, and small metal parts in workshops or laboratories.

2. Assembly assistance

It can assist in assembling lightweight mechanical components by holding parts in position while fastening, aligning, or joining operations are performed.

3. Mechanical testing platform

The arm can be used as a test platform for studying joint motion, torque analysis, load distribution, and mechanical stress behavior in small robotic systems.

4. Precision positioning mechanism

The robotic arm can be used for positioning objects at specific angles or coordinates, making it useful in calibration tasks and small mechanical alignment operations.

5. CNC and automation prototype model

It can serve as a prototype model for understanding the working principles of cnc machines, automated manipulators, and robotic production systems.

6. Packaging and sorting (small components)

The arm can mechanically sort and place small mechanical parts into designated areas, demonstrating automated packaging and sorting principles.

7. Gripping mechanism study

The gripper system can be used to study mechanical gripping forces, gear linkages, and motion transmission in robotic end-effectors.

8. Mechanical motion analysis

The robotic arm can be used to study kinematic chains, degrees of freedom, rotational joints, and link mechanisms in mechanical engineering experiments.

CHAPTER – 8

COST ESTIMATION

8.1 DIRECT MATERIAL COST:

S.NO	DESCRIPTION of COMPONENTS	SPECIFICATIONS	QUANTITY	RATE in Rs	TOTAL in Rs
1	Robotic arm structure(frame + servo motor)	6 DOF Aluminum frame with 4.8–6V servo motors	1	9500	9500/-
2	Esp32 microcontroller	Dual-core 32-bit processor, 240 MHz, 3.3V logic, built-in WI-FI & Bluetooth, 30+ GPIO pins, 4MB flash.	1	600	600/-
3	Lithium ion battery	7.4V nominal (2S), 8.4V full charge, 2000–3000 MAH capacity, rechargeable 18650 cells.	6	100	600/-
4	Buck converter	DC-DC step-down module, 5–40v input, adjustable 6v output, up to 3a current.	1	500	500/-
5	Boost converter	DC-DC step-up module, 2–24V input, adjustable 5V output, up to 2A current	1	50	50/-

6	Jumper wires	22–26 AWG insulated copper wires, male/female connectors, suitable for low-voltage dc connections.	1	200	200/-
7	8.4V charger	8.4V DC output, 1–2A charging current, designed for 2s lithium-ion battery packs.	1	190	190/-
8	TP4056 module	5V input lithium-ion charging module, 1a charging current, overcharge and short-circuit protection.	1	60	60/-
9	Capacitor	4700 µf, 25V electrolytic capacitor for voltage stabilization and spike suppression	2	40	80/-
10	Switch	SPST DC switch, 12V rating, 3–5a current capacity.	1	20	20/-
11	Wooden board	500mm × 250mm	1	200	200/-

TOTAL COST = ₹12,000/-

8.2 Indirect Material Cost -₹200.00

8.3 Labour Cost -₹1000.00

8.4 Fabrication -₹1000.00

8.5 Travel Expenses -₹300.00

8.6 Overhead Cost -₹500.00

TOTAL COST = Direct Material Cost + Indirect Material Cost + Labour Cost + Overhead Cost + Fabrication Cost + Travel Expenses

= 12,000.00 + ₹200.00 + ₹1000.00 + ₹500.00 + ₹1000.00 + ₹300.00

TOTAL COST = 15,000.00

(Fifteen Thousand Only)

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CONCLUSION

The “Design and fabrication of a 6 DOF robotic arm using ESP32” project successfully demonstrates the integration of mechanical design, embedded systems, and wireless communication to create a functional and efficient robotic manipulator. The developed system is capable of performing controlled and precise movements using six servo motors, each representing one degree of freedom. By utilizing PWM signals generated from the ESP32 MICROCONTROLLER, accurate positioning and smooth motion of each joint have been achieved.

One of the major achievements of this project is the implementation of a wireless web-based control interface. This allows the robotic arm to be operated remotely through a mobile phone or computer without requiring additional hardware controllers. Features such as speed adjustment, record and playback functionality and home position initialization enhance the system’s flexibility and automation capability.

The project also provides valuable hands-on experience in robotics, motion control, embedded programming, and power management. It demonstrates how mechanical components and electronic systems must work together in coordination to achieve smooth and stable robotic movement.

Although the robotic arm is designed primarily for lightweight tasks and academic purposes, it lays a strong foundation for further advancements. Future improvements may include the integration of sensors, artificial intelligence, computer vision or industrial-grade actuators to enhance functionality and precision.

In conclusion, this project successfully achieves its objective of designing and developing a compact, affordable and functional robotic arm system. It highlights the practical application of automation principles and provides a stepping stone toward advanced robotic system development.

CHAPTER – 10

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PHOTOGRAPHY